ANGELES UNIVERSITY FOUNDATION
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Final Project

AssistAnt Orchestrator Routing Report

Summary

The Orchestrator receives every request, matches it against markdown routines with regex triggers, delegates work to tiered sub-agents, selects a model provider from a tier-specific chain, and falls back to rule-based execution when no provider is available [1, 2, 3].

Key findings

- Requests are matched against case-insensitive regex triggers; unmatched requests run the unplanned flow and draft a new routine for approval [1, 2].
- Eleven sub-agents are assigned a tier (fast, smart, best, or code); code-tier agents run instantly without a model [1].
- The router walks the tier's provider chain and calls the first provider whose health check returns ok; other states are full, off, nokey, or offline [3].
- Daily caps (e.g., Gemini Flash 20, Groq 1,000) cause providers to be skipped once reached; a 429 marks them exhausted for the day [3].

How it works

- A request arrives at the Orchestrator [1].
- The Orchestrator tests all routine triggers and runs the first matching routine; if none match, it executes the unplanned flow and drafts a new routine for approval [1, 2].
- The routine's steps are delegated to sub-agents; each agent has a tier that determines its model chain [1].
- An agent asks the router for its tier; the router walks the chain—best: Claude Code → Groq gpt-oss-120b → free list → OpenRouter → Ollama; smart: Gemini Flash → Groq gpt-oss-120b → free list → OpenRouter → Ollama → Claude API; fast: Groq gpt-oss-20b → Gemini Flash-Lite → Groq Qwen → free list → OpenRouter → Ollama → Claude API—and selects the first provider with health ok [3].

Risks and limitations

- When all providers are down, only rule-based fallback is available, which may reduce output quality [3].
- Code-tier agents cannot invoke models, limiting them to predefined rule logic [1].
- Incorrect health-check reporting could cause failed provider calls or unnecessary fallbacks [3].
- Routine matching uses simple regex; complex intents may fall to the unplanned flow [1, 2].

References

[1] architecture.md

[2] routines.md

[3] model-router.md

Method: written by a team of AI agents from 3 local documents. Each statement was checked against the source passages (24/24 supported) before it was kept.